MACHINE LEARNING TEAM PROJECT

MIDTERM

Customer Churn Prediction

Research Direction Review & Methodology

Team 소정해

April 7, 2026

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RESEARCH TOPIC & WHY IT MATTERS

Customer Churn Prediction

Machine Learning for Business Retention Strategy

The Problem

5–7x

More expensive to acquire a new customer than retain an existing one

Source: Harvard Business Review

Business & Economic Impact

+25-95%

Profit Increase

"Increasing customer retention rates by just 5% increases profits by 25% to 95%"

Source: Bain & Company

Social Good

- Prevents service loss for vulnerable customers
- Enables personalized retention over blanket pricing
- Promotes fair, data-driven customer management

Literature Review Timeline (2011–2026)

10
Papers Reviewed

2011

Verbeke et al.
Profit-driven
11 Datasets

2019

Ebrah & Elnasir
SVM Comparison
AUC 0.99

2021

Wu et al.
Integrated
Framework

Abdulkadir
DL + Cell2Cell
Resampling

Fujo et al.
Deep-BP-ANN
Lasso

2022

Liu et al.
Hybrid Stacking
Acc 96%

2023

Khattak et al.
BiLSTM-CNN
Deep Learning

2025

Bugajev et al.
Concept Drift
Sliding Window

2026

El Attar et al.
Explainable AI
Ensemble

Manzoor et al.
e-Profits
Business Metric

Profit-Driven

Traditional ML

Deep Learning & Hybrid

Advanced DL

Explainable & Business

Research Evolution: Early Foundations

2011

Verbeke et al.

Profit-Driven Foundation

21 **Algorithms Tested**
Benchmarked across 11 real-life telecom datasets worldwide

6-8 **Variables Needed**
Key finding: Only usage-based core features suffice for accuracy

$ **Profit > Accuracy**
Proposed Maximum Profit (MP) criterion for evaluation

8-YEAR GAP

2019

Ebrah & Elnasir

Traditional ML Comparison

NB | **SVM** | DT | Winner

AUC Performance

Cell2Cell: 0.99 (Best)

IBM: 0.87

Key Churn Signals

- Fiber Optic Internet
- E-Check Payment
- Tenure < 14 Months

Research Evolution: Phase 3 (2021–2022) Deep Learning & Hybrid Era

Wu et al. 2021

Integrated Framework

6 ML Models → Bayesian Analysis → K-means Segment → Targeted Strategy

93.6%
RF Best Accuracy

- 3 Datasets Tested
- Integrated segmentation approach

Abdulkadir 2022

DL Resampling Limits

SMOTE / ROS / RUS

Resampling FAILED
Did not improve DL performance

0.62
Best AUC (Unsampled)

Cell2Cell Dataset (29% Churn)

Fujo et al. 2022

Deep-BP-ANN Optimization

Model	Accuracy
Deep-BP-ANN (Proposed)	88.1%
XGBoost	79.5%
Logistic Regression	76.8%

IBM Telco Dataset

Liu et al. 2022

Hybrid Stacking

SOTA

96%
Accuracy on Orange

Cell2Cell Performance
93.6%

k-medoids + GBT/DT/DL Stacking

Research Evolution: Phase 4–5 Explainability & Business

Phase 4 (2023) Advanced Deep Learning

Composite Deep Learning

Khattak et al., 2023

BiLSTM + CNN → Prediction

Performance Comparison (Accuracy)

Model	Accuracy
BiLSTM-CNN (Proposed)	81%
RF / XGBoost	79%
SVM / RNN	76%
BiLSTM (Standalone)	72%

Kaggle Telco, 7K rows

Phase 5 (2025–2026) Explainability & Business Alignment

40d → 90d

Partial labels predict full churn

Sliding window approach • Addresses concept drift

Bugajev et al., 2025

0.932 AUC

Highest discriminative ability

Explainable ensemble • 3 customer clusters identified • Actionable insights

El Attar & El-Hajj, 2026

e-Profits

Business-aligned metric

Re-ranks models vs AUC/F1 • Integrates CLV & retention probability

Manzoor et al., 2026

Accuracy → **Profit & Value**

Static Data → **Temporal / Drift**

Black-box → **Explainable AI**

Research Gap & Our Goals

CORE RESEARCH QUESTION

"How do behavioral and service quality patterns predict customer churn — and what retention strategies can businesses act on?"

GAPS IDENTIFIED FROM 10 PAPERS

Statistical Metrics Only
Focus on AUC/F1 accuracy, lacking financial alignment.

Static Data
Single-timepoint analysis ignores concept drift over time.

Limited Generalizability
Often validated on a single industry or dataset.

Black-Box Models
High accuracy but insufficient interpretability for business.

Our Response →

OUR RESEARCH GOALS

Aim to Improve Interpretability
Use Feature Importance techniques (like SHAP/XGBoost) to gain actionable insights from the model.

Analyze Behavioral Usage Patterns
Identify key behavioral risk factors using service usage data already present in the dataset.

Try Business-Aligned Evaluation
Attempt to incorporate cost/benefit metrics beyond standard accuracy to measure real business value.

Dataset Candidates

Cell2Cell Telecom

Telecom / Duke University

71,047 ROWS

58 FEATURES

Missing Values: Very Low

Churn Rate: ~50%

U.S. telecom — call quality, billing, complaint history included.

General Churn

General / Kaggle

440,882 ROWS

12 FEATURES

Missing Values: Minor

Churn Rate: ~56%

Industry-agnostic — payment delay, usage frequency, contract type.

IBM Telco Extended

Telecom / IBM Official

7,043 ROWS

33 FEATURES

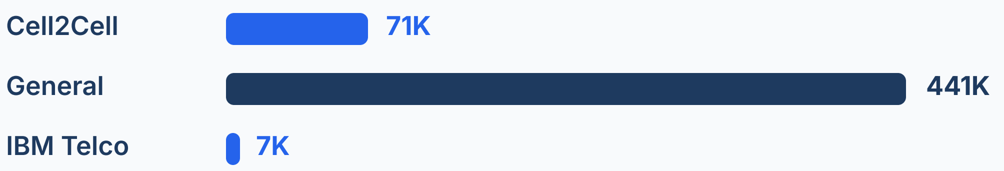
Missing Values: Minor

Churn Rate: 26.5%

California telecom — satisfaction scores, churn reasons included.

Visual Comparison

DATASET SIZE (ROWS)

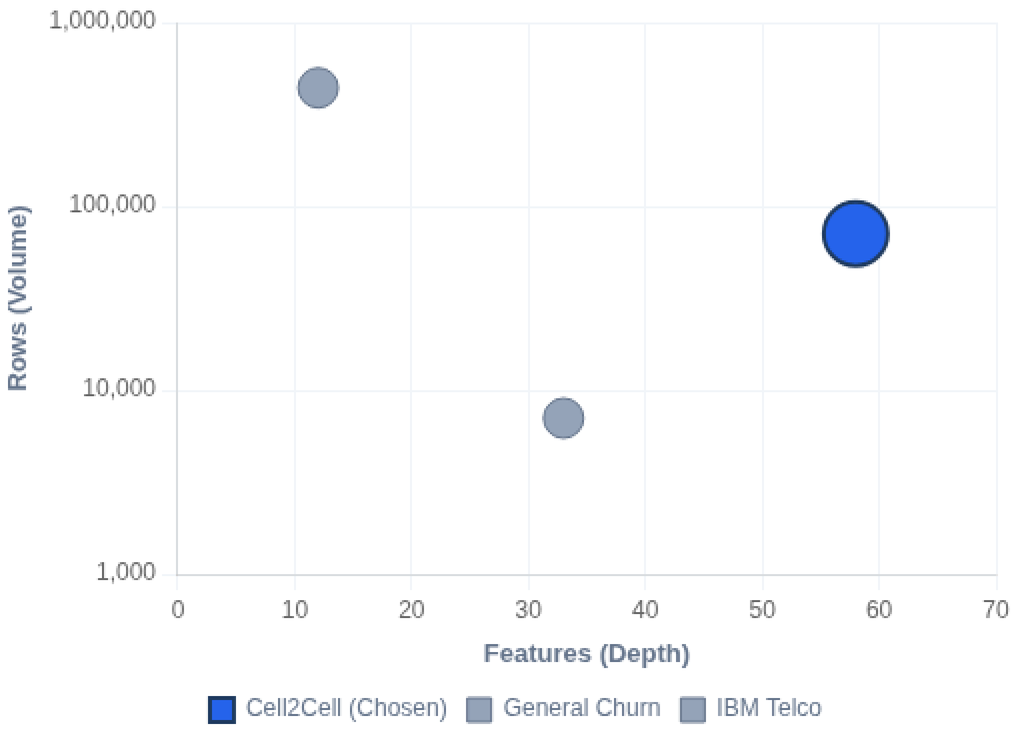


FEATURE RICHNESS

Our Choice: Cell2Cell Telecom

🏛 Duke University Teradata Center

71,047 Rows **58 Features**



Scale + Richness

Optimal balance:
71K rows & 58 vars.

Full Coverage

Covers Service, Finance
& Profile vars.

General Churn Issues

Source Unknown: Unverified origin.
Too Shallow: Only 12 features.

IBM Telco Issues

Too Small: High risk of overfitting.
Features OK but Low Volume.

Data Preprocessing Methodology

```python
# 1) 데이터 처리를 위한 기본 라이브러리 불러오기
import pandas as pd                                # 표 형태 데이터 처리를 위한 pandas 라이브러리
import numpy as np                                 # 수치 계산 및 배열 연산을 위한 numpy 라이브러리

# 2) 학습/검증 데이터 분리를 위한 함수 불러오기
from sklearn.model_selection import train_test_split   # train / test 분할을 위한 함수

# 3) 결측치 대체와 스케일링을 위한 전처리 도구 불러오기
from sklearn.impute import SimpleImputer               # 결측치를 특정 값(여기서는 중앙값)으로 대체하기 위한 도구
from sklearn.preprocessing import RobustScaler         # 이상치에 덜 민감한 스케일링을 수행하기 위한 도구

# 4) CSV 파일 불러오기
df = pd.read_csv("cell2celltrain.csv")                 # 현재 작업 폴더에 있는 데이터셋 CSV 파일을 읽어서 DataFrame으로 저장


# -----------------------------------------------
# STEP 1. 가설에 필요한 변수만 선택하기
# -----------------------------------------------

# 5) 가설 1에서 사용할 핵심 변수명 목록 정의
selected_cols = [
    "Churn",                  # 타겟 변수: 고객 이탈 여부
    "OverageMinutes",         # 가설 H1 핵심 변수: 초과 사용 시간
    "RoamingCalls",           # 가설 H2 핵심 변수: 로밍 통화 수
    "PercChangeRevenues"      # 가설 H3 핵심 변수: 요금 변화율
]

# 6) 원본 데이터프레임에서 필요한 변수만 복사하여 가설 1 전용 데이터셋 생성
hg1_df = df[selected_cols].copy()                     # copy()를 사용해 원본 데이터와 독립적인 복사본 생성


# -----------------------------------------------
# STEP 2. 타겟 변수(Churn)를 숫자로 변환하기
# -----------------------------------------------

# 7) Churn 값이 Yes/No 형태인지 먼저 확인할 수 있도록 고유값 확인
print("Churn unique values:", hg1_df["Churn"].unique())   # 타겟 변수의 실제 고유값 점검
```

Screenshot: Python Preprocessing Pipeline (Pandas + Scikit-Learn)

Dataset Reduction: 71,047 → 약 50,000개 rows.
Original dataset was pre-split into train/test, but test set had no Churn labels, so only train set was used.

Basic Preprocessing Pipeline

1. **Variable Selection:** Selected 4 core variables (Churn, OverageMinutes, RoamingCalls, PercChangeRevenues)
2. **Binary Encoding:** Convert target variable `Churn` (No=0, Yes=1)
3. **Train/Test Split:** 80:20 split with `stratify=y` to maintain class distribution ratio
4. **Imputation:** Fill missing values with **Median** (calculated on Train set only)
5. **Outlier Clipping:** Cap values at 1% (min) and 99% (max) quantiles
6. **Log Transformation:** Apply `np.log1p()` for highly skewed variables (Overage, Roaming)
7. **Scaling:** Apply **RobustScaler** to minimize influence of remaining outliers

Hypotheses in Prior Research

2 / 10

Papers with Explicit Hypotheses

80% used Research Questions or Objectives instead

Wu et al. (2021) — Statistical Hypothesis

T-TEST

- ✕ **H0:** SMOTE does not change classifier performance
- ✓ **H1:** SMOTE significantly changes classifier performance

Bugajev et al. (2025) — Research Hypotheses

★ CORE

- **H1:** Mapping predictions to application windows reduces concept drift impact
- → **H2:** 40-day partial churn labels can predict 90-day full churn via threshold adjustment

** The only paper among 10 to formulate core research hypotheses driving the study.*

8 PAPERS — RESEARCH QUESTIONS / OBJECTIVES

Verbeke et al., 2011	OBJECTIVES
Ebrah & Elnasir, 2019	OBJECTIVES
Abdulkadir, 2022	RESEARCH QUESTIONS
Fujo et al., 2022	OBJECTIVES
Liu et al., 2022	OBJECTIVES
Khattak et al., 2023	RESEARCH QUESTIONS
El Attar & El-Hajj, 2026	OBJECTIVES
Manzoor et al., 2026	CONTRIBUTIONS

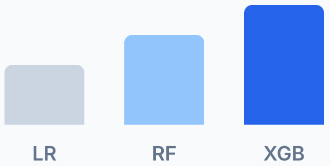
Observation: In Machine Learning / Data Science research, defining **Research Questions (RQs)** and **Objectives** is more common than formulating traditional hypotheses (H1, H2). This is standard practice in the field.

Our Research Hypotheses

Derived from synthesis of 10 prior studies

H1 Ensemble vs. Baseline Performance

"Ensemble models (XGBoost, RF) will achieve significantly higher AUC than the baseline (Logistic Regression)"



↑ Expected AUC Increase

Compare AUC across 3 models

Evidence: Fujo 2022 · Liu 2022 · El Attar 2026

H2 Impact of Feature Engineering

"Engineered features (Revenue/Month, Call Drop Rate) will improve prediction vs. using original variables only"

Original Vars → + Engineered

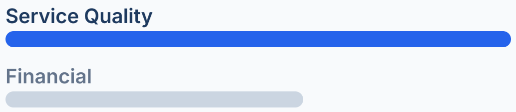
Better Accuracy

With vs. Without engineered features

Evidence: Verbeke 2011 · Fujo 2022

H3 Service Quality Importance

"Service quality variables (dropped calls, care calls) will show higher feature importance than financial variables"



Higher Ranking

Feature importance ranking

Evidence: Ebrah 2019 · Wu 2021 · El Attar 2026

Variable Design & ML Approach

Variable Design

TARGET VARIABLE (Y)

Y = Churn (0: Retain, 1: Leave)

Binary Classification Problem

SERVICE QUALITY

- DroppedCalls
- BlockedCalls
- CustomerCareCalls
- ServiceFailures

FINANCIAL

- MonthlyRevenue
- OverageMinutes
- TotalCharges
- PaymentDelay

LOYALTY

- MonthsInService
- ContractType
- RenewalStatus
- ActiveMember

ENGINEERED FEATURES

- Revenue per Month
- Call Drop Rate
- Usage Trend

ML Approach Pipeline

BASELINE

Logistic Regression

Interpretable linear benchmark to establish performance floor.

↓

ENSEMBLE

Random Forest

Handles non-linearity, robust performance via bagging strategy.

↓

BOOSTING

XGBoost

Gradient boosting, highest accuracy for tabular data.

↓

Accuracy | Precision | Recall | F1-Score | **AUC-ROC**

Research Methodology Framework

End-to-End Workflow: From Data Acquisition to Strategic Insights

Data Collection

Cell2Cell Telecom Dataset
71,047 Rows | 58 Features

Preprocessing

- Binary Encoding
- Train/Test Split (80:20)
- Median Imputation
- Robust Scaling

Exploratory Analysis

- Distribution Plots
- Correlation Matrix
- Churn Rate Analysis
- Outlier Detection

Model Training

Intensive Phase

Training & Tuning with GridSearchCV

Logistic Regression | Random Forest | XGBoost

Evaluation

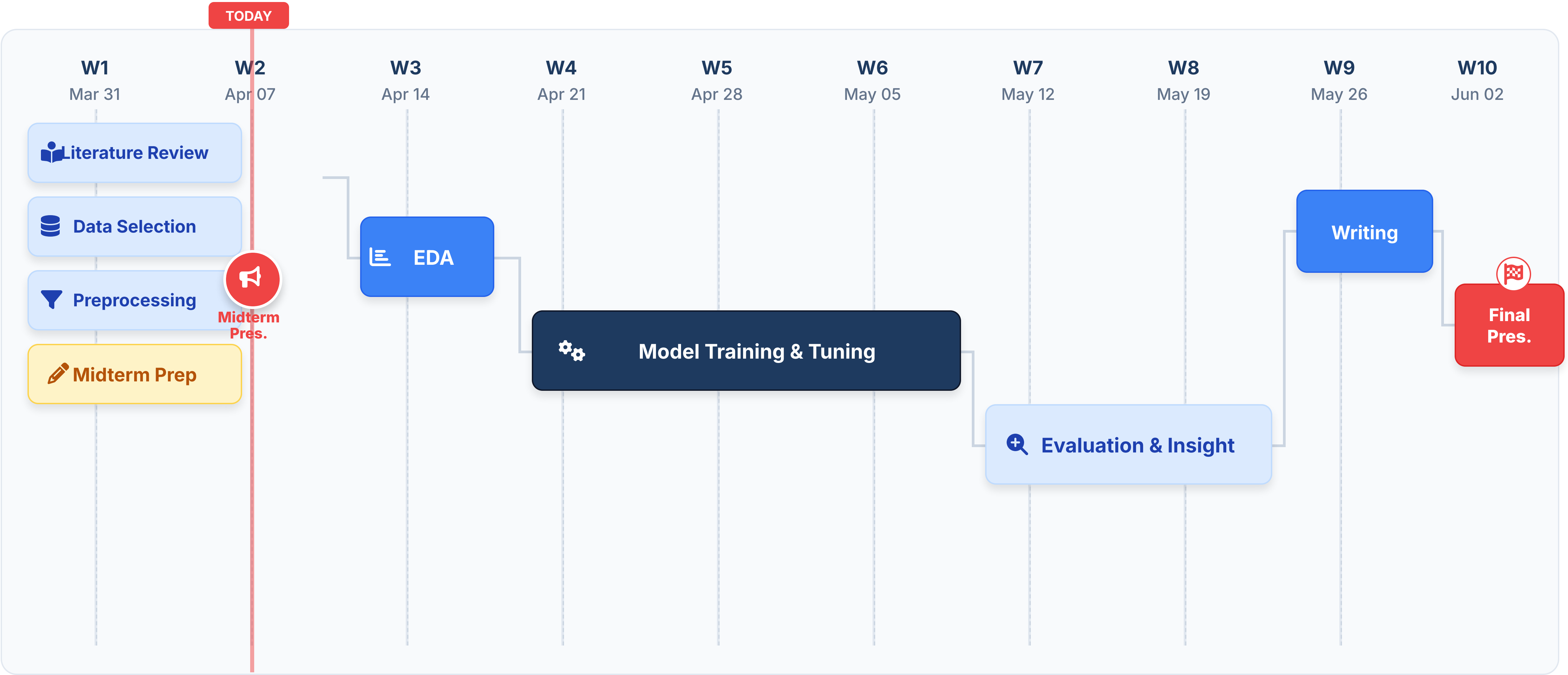
5-Fold Cross Validation
AUC-ROC, Accuracy, F1

Insight & Strategy

Feature Importance Analysis
Retention Recommendations

10-Week Execution Timeline

Project Schedule: March 31 – June 2, 2026



Questions?

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